Customer Service Request Analysis

Task that performed

1. import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import kruskal

2. df = pd.read_csv("311_Service_Requests.csv")

3. print(df.shape) # Shape of dataset
print(df.columns) # Column names
print(df.head()) # First 5 rows
print(df.info()) # Dataset info

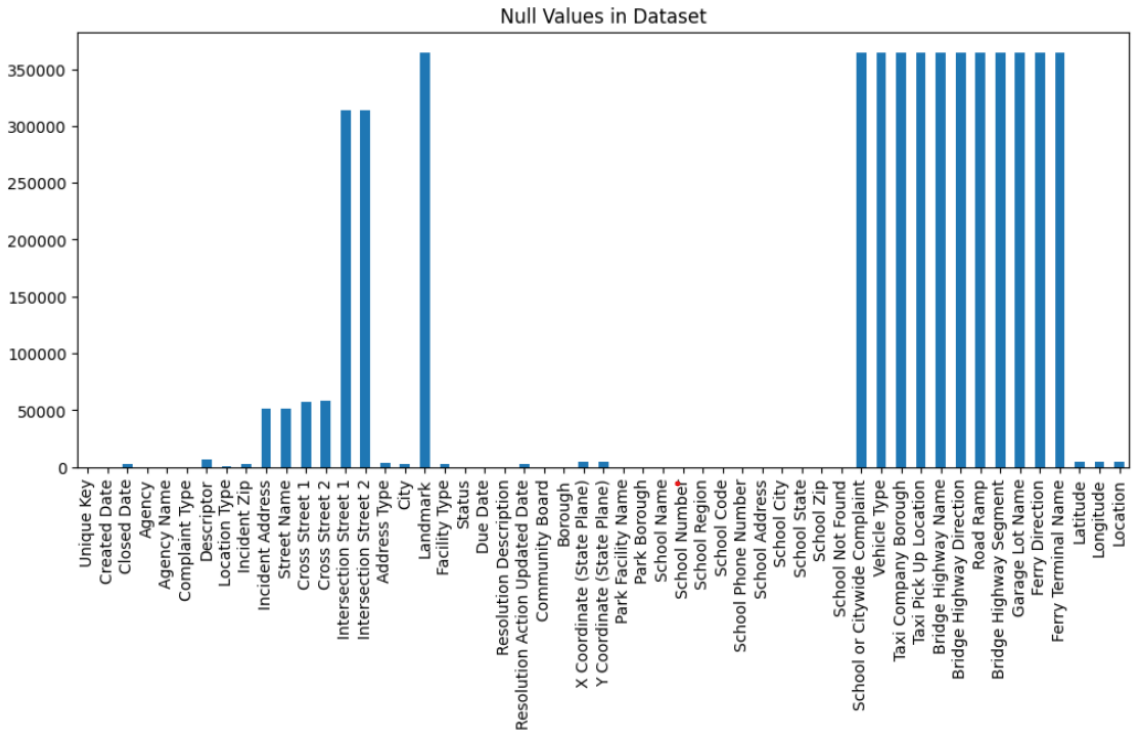
```
(364558, 53)
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
   Unique Key            Created Date       Closed Date Agency  \
0    32310363  12/31/2015 11:59:45 PM  01-01-2016 00:55   NYPD
1    32309934  12/31/2015 11:59:44 PM  01-01-2016 01:26   NYPD
2    32309159  12/31/2015 11:59:29 PM  01-01-2016 04:51   NYPD
3    32305098  12/31/2015 11:57:46 PM  01-01-2016 07:43   NYPD
4    32306529  12/31/2015 11:56:58 PM  01-01-2016 03:24   NYPD

                      Agency Name           Complaint Type  \
0  New York City Police Department  Noise - Street/Sidewalk
1  New York City Police Department         Blocked Driveway
2  New York City Police Department         Blocked Driveway
3  New York City Police Department          Illegal Parking
4  New York City Police Department          Illegal Parking

                         Descriptor    Location Type  Incident Zip  \
0              Loud Music/Party  Street/Sidewalk       10034.0
1                     No Access  Street/Sidewalk       11105.0
2                     No Access  Street/Sidewalk       10458.0
3  Commercial Overnight Parking  Street/Sidewalk       10461.0
4              Blocked Sidewalk  Street/Sidewalk       11373.0

         Incident Address  ... Bridge Highway Name Bridge Highway Direction  \
0    71 VERMILYEA AVENUE  ...                  NaN                       NaN
1        27-07 23 AVENUE  ...                  NaN                       NaN
2  2897 VALENTINE AVENUE  ...                  NaN                       NaN
3    2940 BAISLEY AVENUE  ...                  NaN                       NaN
4          87-14 57 ROAD  ...                  NaN                       NaN

  Road Ramp Bridge Highway Segment Garage Lot Name Ferry Direction  \
0       NaN                    NaN             NaN             NaN
1       NaN                    NaN             NaN             NaN
2       NaN                    NaN             NaN             NaN
3       NaN                    NaN             NaN             NaN
4       NaN                    NaN             NaN             NaN

  Ferry Terminal Name   Latitude  Longitude  \
0                 NaN  40.865682 -73.923501
1                 NaN  40.775945 -73.915094
2                 NaN  40.870325 -73.888525
3                 NaN  40.835994 -73.828379
4                 NaN  40.733060 -73.874170

                                     Location
0   (40.86568153633767, -73.92350095571744)
1  (40.775945312321085, -73.91509393898605)
2  (40.870324522111424, -73.88852464418646)
3   (40.83599404683083, -73.82837939584206)
4  (40.733059618956815, -73.87416975810375)
```

```
[5 rows x 53 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 364558 entries, 0 to 364557
Data columns (total 53 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   Unique Key                      364558 non-null  int64
 1   Created Date                    364558 non-null  object
 2   Closed Date                     362177 non-null  object
 3   Agency                          364558 non-null  object
 4   Agency Name                     364558 non-null  object
 5   Complaint Type                  364558 non-null  object
 6   Descriptor                      358057 non-null  object
 7   Location Type                   364425 non-null  object
 8   Incident Zip                    361560 non-null  float64
 9   Incident Address                312859 non-null  object
 10  Street Name                     312859 non-null  object
 11  Cross Street 1                  307370 non-null  object
 12  Cross Street 2                  306753 non-null  object
 13  Intersection Street 1           51120 non-null   object
 14  Intersection Street 2           50512 non-null   object
 15  Address Type                    361306 non-null  object
 16  City                            361561 non-null  object
 17  Landmark                        375 non-null     object
 18  Facility Type                   362169 non-null  object
 19  Status                          364558 non-null  object
 20  Due Date                        364555 non-null  object
 21  Resolution Description          364558 non-null  object
 22  Resolution Action Updated Date  362156 non-null  object
 23  Community Board                 364558 non-null  object
 24  Borough                         364558 non-null  object
 25  X Coordinate (State Plane)      360528 non-null  float64
 26  Y Coordinate (State Plane)      360528 non-null  float64
 27  Park Facility Name              364558 non-null  object
 28  Park Borough                    364558 non-null  object
 29  School Name                     364558 non-null  object
 30  School Number                   364558 non-null  object
 31  School Region                   364557 non-null  object
 32  School Code                     364557 non-null  object
 33  School Phone Number             364558 non-null  object
 34  School Address                  364558 non-null  object
 35  School City                     364558 non-null  object
 36  School State                    364558 non-null  object
 37  School Zip                      364557 non-null  object
 38  School Not Found                364558 non-null  object
 39  School or Citywide Complaint    0 non-null       float64
 40  Vehicle Type                    0 non-null       float64
 41  Taxi Company Borough            0 non-null       float64
 42  Taxi Pick Up Location           0 non-null       float64
 43  Bridge Highway Name             297 non-null     object
 44  Bridge Highway Direction        297 non-null     object
 45  Road Ramp                       262 non-null     object
 46  Bridge Highway Segment          262 non-null     object
 47  Garage Lot Name                 0 non-null       float64
 48  Ferry Direction                 1 non-null       object
 49  Ferry Terminal Name             2 non-null       object
 50  Latitude                        360528 non-null  float64
 51  Longitude                       360528 non-null  float64
 52  Location                        360528 non-null  object
dtypes: float64(10), int64(1), object(42)
memory usage: 147.4+ MB
None
```

4. print(df.isnull().sum())

```
Unique Key                            0
Created Date                          0
Closed Date                        2381
Agency                                0
Agency Name                           0
Complaint Type                        0
Descriptor                         6501
Location Type                       133
Incident Zip                       2998
Incident Address                  51699
Street Name                       51699
Cross Street 1                    57188
Cross Street 2                    57805
Intersection Street 1            313438
Intersection Street 2            314046
Address Type                       3252
City                               2997
Landmark                         364183
Facility Type                      2389
Status                                0
Due Date                              3
Resolution Description                0
Resolution Action Updated Date     2402
Community Board                       0
Borough                               0
X Coordinate (State Plane)         4030
Y Coordinate (State Plane)         4030
Park Facility Name                    0
Park Borough                          0
School Name                           0
School Number                         0
School Region                         1
School Code                           1
School Phone Number                   0
School Address                        0
School City                           0
School State                          0
School Zip                            1
School Not Found                      0
School or Citywide Complaint     364558
Vehicle Type                     364558
Taxi Company Borough             364558
Taxi Pick Up Location            364558
Bridge Highway Name              364261
Bridge Highway Direction         364261
Road Ramp                        364296
Bridge Highway Segment           364296
Garage Lot Name                  364558
Ferry Direction                  364557
Ferry Terminal Name              364556
Latitude                           4030
Longitude                          4030
Location                           4030
dtype: int64
```

5. df.isnull().sum().plot(kind='bar', figsize=(12,5))
   plt.title("Null Values in Dataset




```
plt.show()
```

6. df = df[df['Closed Date'].notnull()]

7. import pandas as pd

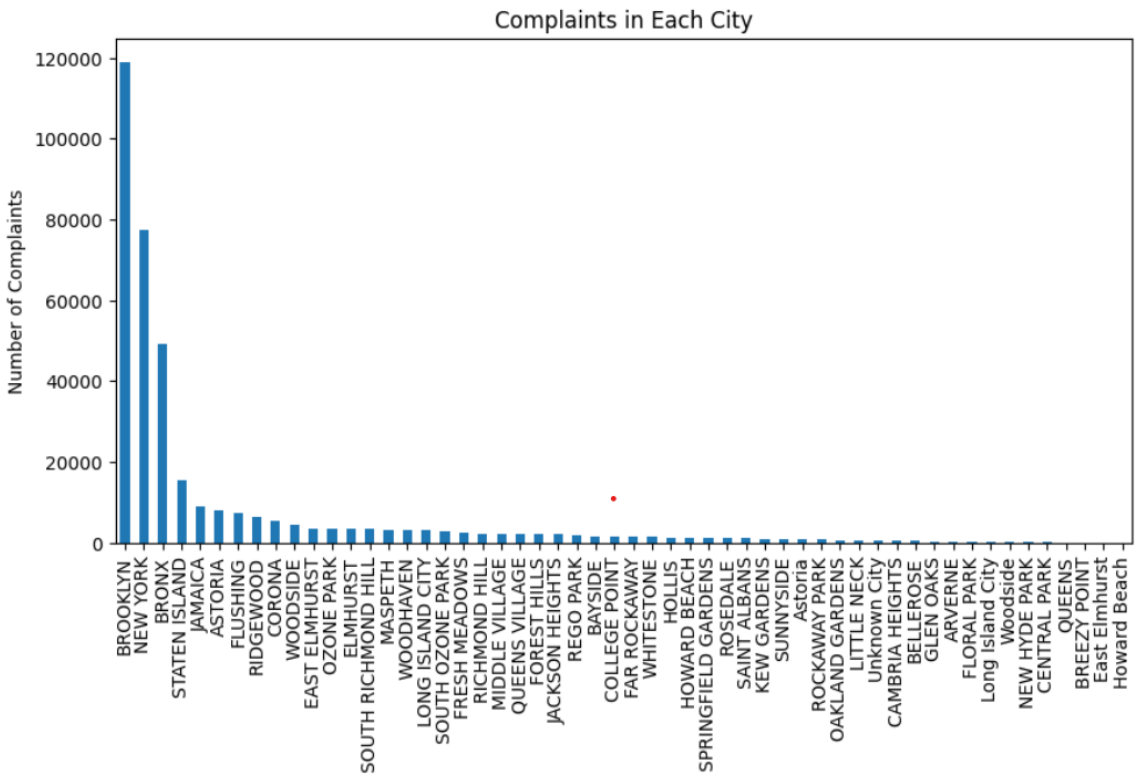
```
df['Created Date'] = pd.to_datetime(df['Created Date'], format="%m/%d/%Y %I:%M:%S %p")
df['Closed Date'] = pd.to_datetime(df['Closed Date'], format='mixed')
```

8. df['Created Date'] = pd.to_datetime(df['Created Date'])
   df['Closed Date'] = pd.to_datetime(df['Closed Date'])

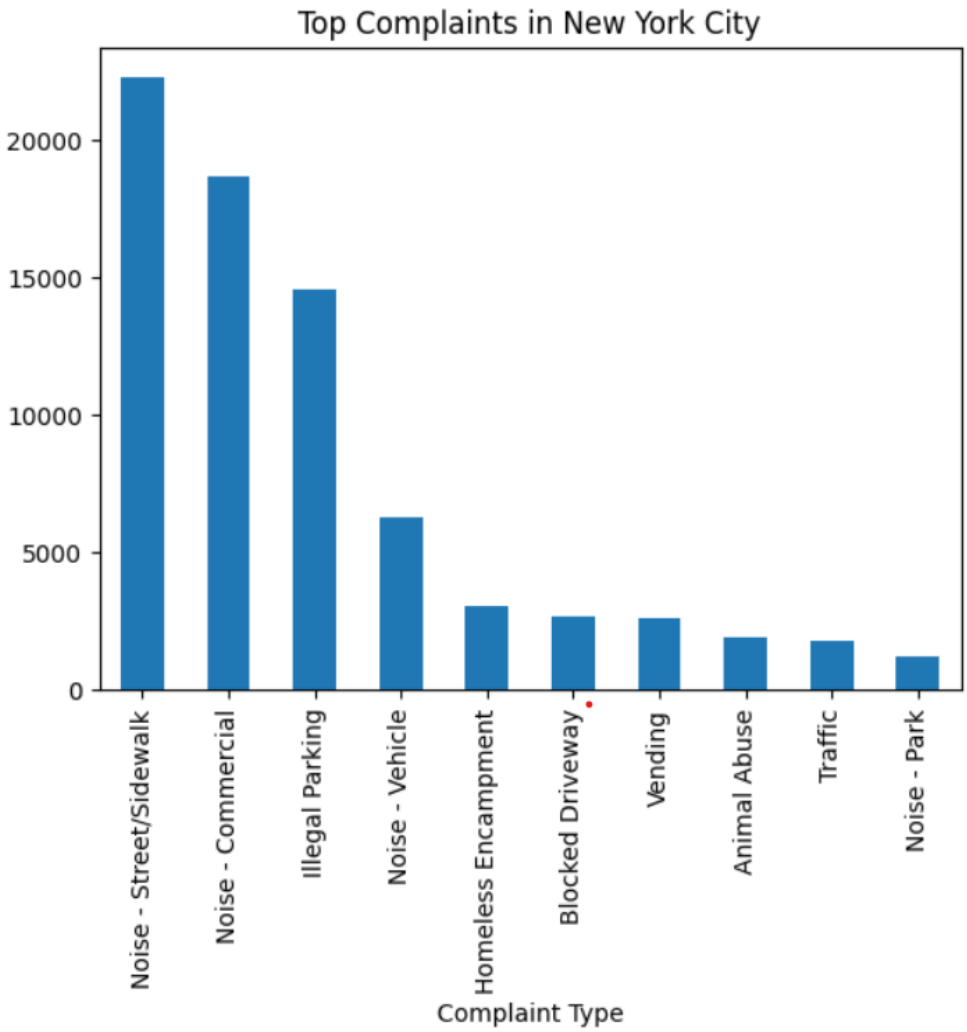
9. df['Request_Closing_Time'] = df['Closed Date'] - df['Created Date']
10. df['Request_Closing_Time'] = df['Request_Closing_Time'].dt.total_seconds()
11. df['City'].fillna("Unknown City", inplace=True)
12. print(df[['Complaint Type','City']].isnull().sum())

```
Complaint Type    0
City              0
dtype: int64
```

13. plt.figure(figsize=(10,5))
    df['City'].value_counts().plot(kind='bar')
    plt.title("Complaints in Each City")
    plt.xlabel("City")
    plt.ylabel("Number of Complaints")
    plt.show()



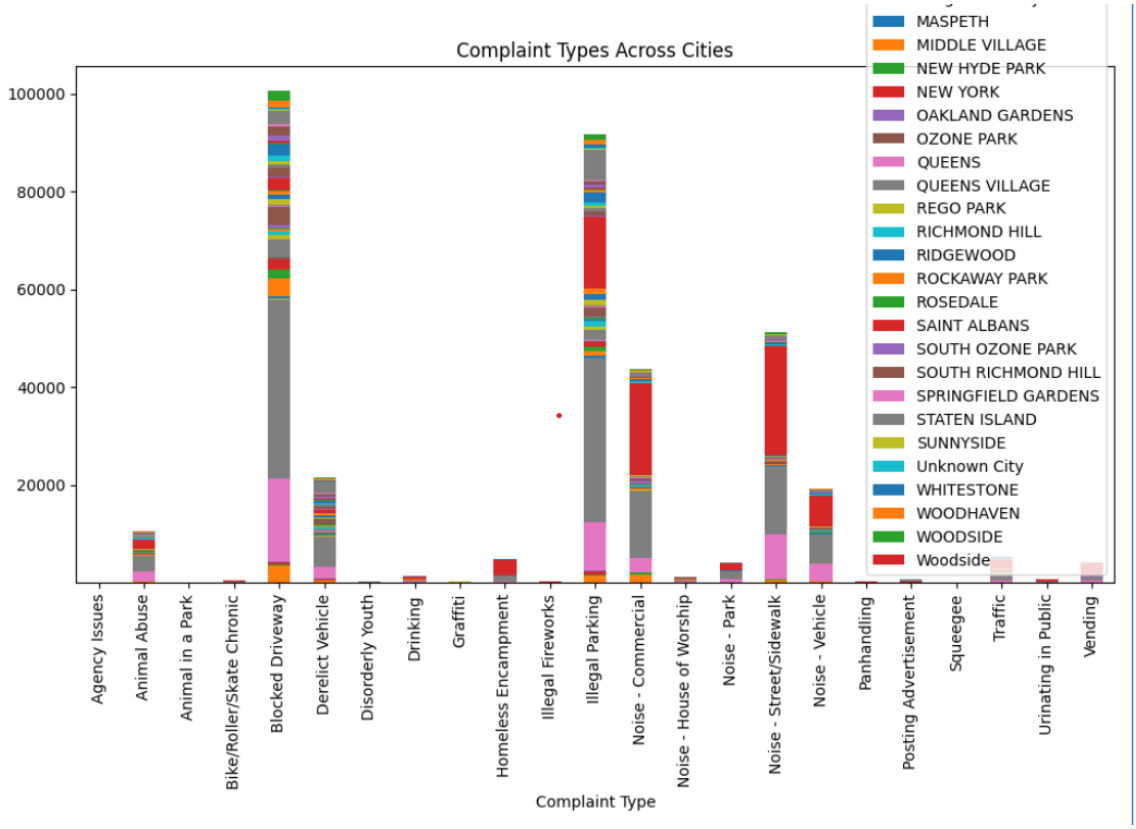
14. nyc = df[df['City'] == 'NEW YORK']
    nyc['Complaint Type'].value_counts().head(10).plot(kind='bar')
    plt.title("Top Complaints in New York City")
    plt.show()



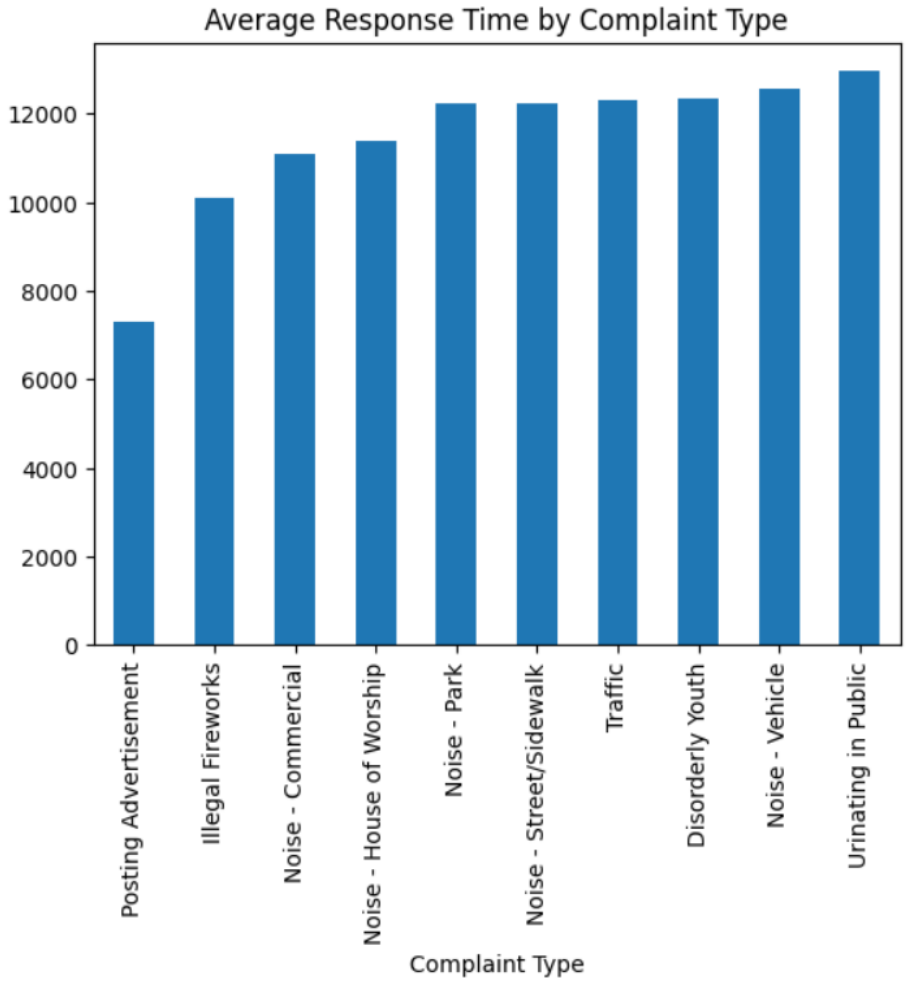
15. df_new = pd.crosstab(df['Complaint Type'], df['City'])
    print(df_new.head())

```
City                        ARVERNE  ASTORIA  Astoria  BAYSIDE  BELLEROSE  \
Complaint Type
Agency Issues                     0        0        0        0          0
Animal Abuse                     46      170        0       53         15
Animal in a Park                  0        0        0        0          0
Bike/Roller/Skate Chronic         0       16        0        0          1
Blocked Driveway                 50     3436      159      514        138

City                        BREEZY POINT  BRONX  BROOKLYN  CAMBRIA HEIGHTS  \
Complaint Type
Agency Issues                           0      0         0                0
Animal Abuse                            2   1971      3191               15
Animal in a Park                        0      0         0                0
Bike/Roller/Skate Chronic               0     22       124                0
Blocked Driveway                        3  17062     36445              177

City                        CENTRAL PARK  ...  SOUTH OZONE PARK  \
Complaint Type                            ...
Agency Issues                           0  ...                 0
Animal Abuse                            0  ...                74
Animal in a Park                        0  ...                 0
Bike/Roller/Skate Chronic               0  ...                 1
Blocked Driveway                        0  ...              1202

City                        SOUTH RICHMOND HILL  SPRINGFIELD GARDENS  \
Complaint Type
Agency Issues                                 0                    0
Animal Abuse                                 40                   42
Animal in a Park                              0                    0
Bike/Roller/Skate Chronic                     1                    0
Blocked Driveway                           1946                  330

City                        STATEN ISLAND  SUNNYSIDE  Unknown City  WHITESTONE  \
Complaint Type
Agency Issues                           0          0             8           0
Animal Abuse                          786         40             1          43
Animal in a Park                        0          0             0           0
Bike/Roller/Skate Chronic              10          2             2           4
Blocked Driveway                     2845        278            86         279

City                        WOODHAVEN  WOODSIDE  Woodside
Complaint Type
Agency Issues                       0         0         0
Animal Abuse                       57       111         0
Animal in a Park                    0         0         0
Bike/Roller/Skate Chronic           2         5         0
Blocked Driveway                 1363      2038        27

[5 rows x 54 columns]
```

16. df_new.plot(kind='bar', stacked=True, figsize=(12,6))
    plt.title("Complaint Types Across Cities")
    plt.show()



17. avg_time = df.groupby('Complaint Type')['Request_Closing_Time'].mean()
    avg_time.sort_values().head(10).plot(kind='bar')
    plt.title("Average Response Time by Complaint Type")
    plt.show()



18. groups = []
    for complaint in df['Complaint Type'].unique():

groups.append(df[df['Complaint Type']==complaint]['Request_Closing_Time'])

19. stat, p = kruskal(*groups)
    print("Kruskal Test Statistic:", stat)
    print("P Value:", p)

```
Kruskal Test Statistic: 11988.190080116212
P Value: 0.0
```

20. alpha = 0.05
    if p < alpha:
    print("Reject Null Hypothesis")
    else:
    print("Fail to Reject Null Hypothesis")

```
Reject Null Hypothesis
```

21. print("Most common complaint:", df['Complaint Type'].value_counts().idxmax())
    print("City with highest complaints:", df['City'].value_counts().idxmax())

```
Most common complaint: Blocked Driveway
City with highest complaints: BROOKLYN
```